

ESSMHomeLab Operations and Readiness Overview

Infrastructure, Services, Monitoring, and Operational Readiness

PUBLIC PORTFOLIO EDITION
Sanitized operational documentation

Public portfolio edition: This document describes the design, administration, monitoring, and operational practices used in ESSMHomeLab. Exact addresses, account names, credentials, private keys, storage paths, share names, and sensitive configuration values are intentionally excluded.

Executive Summary

ESSMHomeLab is a working multi-platform environment built for hands-on systems administration, technical support, virtualization, networking, application support, automation, and operational documentation. It combines three Proxmox VE hosts, Windows Server Active Directory and DNS, Ubuntu and Debian servers, Docker applications, centralized storage, Linux and Windows administration workstations, and custom health-reporting tools.

The environment is managed as a small production-style infrastructure platform. Systems are assigned defined roles, dependencies are documented, expected network services are tested, pending reboots are tracked, and application readiness is verified beyond simple host availability.

Environment at a Glance

Area	Implementation
Virtualization	Three independently managed Proxmox VE hosts
Identity and DNS	Windows Server 2022 with Active Directory Domain Services and Microsoft DNS
Server platforms	Ubuntu Server, Debian, and Linux Mint
Application platform	Docker and Docker Compose
Storage	Centralized NFS and Samba services with separate primary and backup storage functions
Administration	Windows 11 with WSL2 and a Linux Mint administrative workstation
Automation	Bash, PowerShell, SSH keys, host aliases, ping tests, and TCP service checks
Monitoring	Consolidated availability, operating-system health, uptime, service-port, and reboot reporting
Network	Private IPv4 LAN with reserved infrastructure addressing and internal DNS

Architecture and Workload Placement

The workload design separates infrastructure services from applications and end-user systems. This makes troubleshooting easier and reduces the impact of maintenance on unrelated services.

Virtualization layer	Primary workloads
Proxmox node 1	Centralized storage and the primary Docker application platform
Proxmox node 2	Active Directory/DNS, security and monitoring, and a multipurpose infrastructure server
Proxmox node 3	Mattermost, Gramps Web, and a Linux learning workstation
Physical Debian host	Jellyfin media services
Administrative systems	Windows/WSL primary console and Linux Mint secondary console

Managed Entity Summary

Entity	Primary use	Operational focus
Proxmox node 1	Hosts storage and Docker workloads	Host resources, storage, Proxmox services, and guest state
Proxmox node 2	Hosts identity and infrastructure workloads	Host resources, Proxmox services, and guest readiness
Proxmox node 3	Hosts collaboration, genealogy, and learning workloads	Memory pressure, VM state, and local storage
Windows domain controller	AD DS, DNS, Kerberos, LDAP, SMB, and remote administration	Directory services, DNS accuracy, authentication, event logs, and diagnostics
Security/monitoring server	Platform for current and future security services	Failed units, resources, authentication logs, and security-tool health
Central storage server	NFS and Samba file services	Mounted filesystems, free capacity, exports, shares, permissions, and restore readiness
Docker application host	Runs infrastructure and business applications	Docker engine, containers, restart counts, logs, certificates, and endpoints
Mattermost server	Team collaboration and messaging	Application process, API, authentication, message delivery, and logs
Infrastructure utility server	File, identity, monitoring, and guest-management support	SSH, Samba, SSSD, monitoring services, resources, and updates
Gramps Web server	Genealogy application and background workers	Containers, application data, authentication, search, media, and backups
Jellyfin server	Media library and playback	Container health, media mounts, library scans, playback, and transcoding
Linux learning workstation	Technical and theology research environment	Desktop login, applications, updates, resources, and data access
Linux admin workstation	Secondary command center	SSH keys, management tools, shared storage, and target connectivity
Windows/WSL admin workstation	Primary command center and reporting platform	OpenSSH, WSL2, scripts, PowerShell integration, and remote access

Service Portfolio

The environment includes both infrastructure services and user-facing applications.

Service category	Technologies and uses	Verification approach
Virtualization	Proxmox VE management and VM lifecycle	Confirm management interface, storage, Proxmox services, and expected guests
Identity	Active Directory, Kerberos, and LDAP	Resolve internal names, authenticate a domain user, and run directory diagnostics
Name services	Microsoft DNS	Query internal records directly and confirm clients use the correct resolver
File services	NFS and Samba	Enumerate exports/shares and complete client-side create, read, and delete tests
Reverse proxy and TLS	Nginx Proxy Manager	Confirm routes, certificates, and the administrative interface
Documentation	BookStack	Test login, search, and page retrieval
Document management	Paperless-ngx	Test login, document search, and retrieval
Network management	UniFi Network Application	Confirm the controller loads and managed devices report current state
Container administration	Portainer	Confirm login and inspect environments, containers, and health status
Performance monitoring	Netdata and Prometheus	Confirm current metrics and healthy collection targets
Collaboration	Mattermost	Test API response, login, and message delivery
Genealogy	Gramps Web	Test login, tree access, search, media, and background jobs
Media	Jellyfin	Test the health endpoint, library access, and client playback
Dashboard	Homepage	Confirm the dashboard loads and links reach their intended services

Expected Network Services

The monitoring inventory checks standard infrastructure ports and application-specific listeners. The public documentation records service types without exposing the private address map.

Service	Typical TCP port	Purpose
SSH	22	Secure remote administration
DNS	53	Internal name resolution
Kerberos	88	Domain authentication
LDAP	389	Directory access
SMB/Samba	445	Windows-compatible file sharing
NFS	2049	Linux file sharing
RDP	3389	Windows remote administration
Proxmox HTTPS	8006	Virtualization management
HTTP/HTTPS	80/443	Web applications and reverse proxy services
Mattermost	8065	Collaboration application
Jellyfin	Custom internal port	Media application

Service	Typical TCP port	Purpose
Application administration	Restricted internal ports	Container, network, proxy, and monitoring consoles

An open TCP port confirms that a listener accepted a connection. It does not prove that authentication, data access, background processing, or the normal user workflow is functioning.

ESSM Monitoring and Automation

Custom shell functions provide a consistent operational view from the primary administration workstation.

Command	Function	Expected healthy result
essmstatus	Tests network availability across the managed inventory	Every expected system reports ONLINE
essmhealth	Collects platform, version, uptime, SSH health, and pending-reboot state	Every system reports HEALTHY; reboot state is NO
essmports	Tests registered TCP service listeners	Every expected listener reports OPEN
essmhealthstatus	Produces the combined availability and detailed-health report	Status and health counts match the managed inventory
essmhelp	Lists available ESSM commands and configured management targets	Current commands and targets are displayed

The scripts use multiple ping attempts and longer or retried SSH connections to reduce false OFFLINE and NO SSH results caused by initial address-resolution or neighbor-discovery delays.

Readiness Model

- 1 Reachability: Confirm the system responds to repeated network probes.
- 2 Port availability: Confirm each expected TCP listener accepts a connection.
- 3 Service health: Verify the required systemd unit, Windows service, VM process, or container is active and stable.
- 4 Functional readiness: Complete the real workload, such as authentication, DNS lookup, file access, message delivery, document retrieval, genealogy search, or media playback.

Report Interpretation

Result	Meaning	Required response
ONLINE	The system replied during the network test	Continue with service and application checks
OFFLINE	No reply was received during the polling window	Retry and test TCP, address resolution, VM state, NIC, and host state
HEALTHY	The remote health query completed successfully	Review reboot and service-specific status
NO SSH	SSH connectivity or authentication failed	Test port 22, username, key, host key, SSH service, and firewall
REBOOT YES	The operating system reports a pending reboot	Schedule a controlled reboot and verify recovery
PORT OPEN	A TCP listener accepted the connection	Perform the application-level functional test

Operational Validation Procedures

Daily or On-Demand Check

- 1 Run `essmhealthstatus` and confirm that the network and health counts match the current inventory.
- 2 Confirm every expected system is `ONLINE` and `HEALTHY`.
- 3 Confirm every reboot indicator is `N0`.
- 4 Run `essmports` and confirm every registered service is `OPEN`.
- 5 Perform functional checks for critical services, including DNS, authentication, storage, collaboration, genealogy, media, and the Docker application platform.
- 6 Investigate any `OFFLINE`, `N0` `SSH`, `REBOOT YES`, or `CLOSED` result before making unrelated changes.
- 7 Record material failures, corrections, address changes, and service changes.

Incident Triage

- 1 Repeat the failed check to rule out a transient packet-loss or neighbor-discovery event.
- 2 Compare network, TCP-port, SSH, process, container, and application results to identify the failing layer.
- 3 If a virtualization management address fails, test its guest systems before declaring the host unavailable.
- 4 For SSH failures, verify port 22, the expected username, authorized key, host key, SSH service, and firewall.
- 5 For an open application port with a failed user workflow, inspect service or container logs and dependencies.
- 6 Apply the least disruptive correction and rerun both the technical and functional tests.

Controlled Reboot and Recovery

- Reboot only one dependent system at a time.
- Verify backups, file transfers, and long-running jobs before rebooting storage or application systems.
- Wait for network and SSH access to return.
- Confirm the new uptime and verify that the reboot-required indicator cleared.
- Validate failed-service state, storage mounts, containers, application endpoints, authentication, and the normal user workflow.
- Rerun `essmhealthstatus` and `essmports` after planned maintenance is complete.

Hypervisors are rebooted only during a planned maintenance window after guests are safely stopped or migrated. The domain controller is handled separately so DNS and authentication can be verified immediately after recovery.

Storage, Security, and Change Control

Storage and Data Protection

- Monitor free capacity and filesystem errors on primary, backup, and virtualization storage.
- Test NFS and Samba from a client system, not only from the storage server.
- Review backup completion and perform periodic test restores.

- Document retention, encryption, destination, and restore ownership for critical datasets.
- Verify dependent mounts and applications after storage maintenance.

Access and Security

- Use key-based SSH from approved administrative workstations.
- Restrict management consoles to the trusted LAN or a secured VPN.
- Review authorized keys, privileged group membership, and firewall rules regularly.
- Keep credentials, tokens, private keys, and recovery information outside the public repository.
- Remove unused services and close ports that are not part of the expected-service inventory.
- Review elevated NFS export options and limit them to explicitly trusted clients.

Maintenance Cadence

Frequency	Review
Daily or on demand	Consolidated health report, critical application functions, and pending reboots
Weekly	Expected ports, failed services, disk capacity, container health, backups, and updates
Monthly	Restore test, account/key review, certificates, inventory, and documentation reconciliation
After every material change	Script syntax, health and port reports, dependencies, service inventory, and change record

Verified Historical Baseline

The September 5, 2026 operational baseline recorded:

- 14 managed systems online
- 14 detailed health checks completed successfully
- 28 combined polling operations completed
- All registered service listeners open
- No pending reboot indicators after planned maintenance

This baseline is evidence of a successful test at that time, not a guarantee of current state. Current command output remains the authoritative operational source.

Skills Demonstrated

- Proxmox VE installation, administration, storage, and VM operations
- Windows Server 2022, Active Directory, DNS, Kerberos, LDAP, SMB, and RDP
- Ubuntu, Debian, and Linux Mint administration
- Docker and Docker Compose deployment and troubleshooting
- NFS and Samba storage administration
- Bash and PowerShell automation
- SSH key management and cross-platform remote administration
- TCP/IP, DNS, port, process, and application-layer troubleshooting
- Monitoring, operational readiness testing, controlled maintenance, recovery validation, runbook development, change control, and security-minded documentation